

# An independent, Lean-verified proof of Written on the Wall II Conjecture 100

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Proof search and formalization assisted by Claude (Anthropic)

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## Abstract

Conjecture 100 of DeLaViña’s *Written on the Wall II* (WOWII), a list of conjectures produced by the program Graffiti.pc, states that every connected graph  $G$  on at least two vertices satisfies  $\alpha(G) \leq \lceil (\max_v l(v) + \frac{1}{2} \text{length}(\overline{G}))/2 \rceil$ , where  $l(v)$  is the independence number of the neighbourhood of  $v$  and  $\text{length}(\overline{G})$  is the Euclidean norm of the degree sequence of the complement. **This is not the first proof.** The conjecture was proved by Kias Henry [2] (Zenodo, August 2026; the proof was reviewed by E. DeLaViña, who marked the conjecture as true), and a Lean formalization by the GitHub user vatsj was submitted to the **formal-conjectures** repository as pull request #5221 [3]. We give an independent argument that does not use the Cauchy–Schwarz or AM–GM inequalities: after elementary bounds on complement degrees and a double count, everything reduces to the positivity of one explicit bivariate polynomial  $P(k, t)$  on the positive integers. We also describe a self-contained Lean 4 / Mathlib formalization of this argument, checked against the unchanged statement in **formal-conjectures**. We completed both the argument and the formalization before learning of the earlier work. Our formalization is about 220 lines with three helper lemmas, compared with the roughly 330 lines and nine helper lemmas reported in pull request #5221.

## 1 Introduction

Graffiti.pc is a conjecture-generating program written by Ermelinda DeLaViña; its output is collected in the online list *Written on the Wall II* (WOWII) [1]. Google DeepMind’s **formal-conjectures** repository [4] contains Lean 4 statements of many WOWII conjectures, each tagged with a status. Conjecture 100 concerns the independence number, local independence numbers, and the degree sequence of the complement graph.

**Prior work.** We state plainly that the result in this note is *already known*:

- Kias Henry, *A Proof of Written on the Wall II Conjecture 100*, Zenodo, 13 August 2026, doi:10.5281/zenodo.21914031 [2]. Henry’s proof uses the Cauchy–Schwarz and AM–GM inequalities. It was reviewed by E. DeLaViña, who marked the conjecture as true on the WOWII page.
- A Lean proof by the GitHub user vatsj, submitted as pull request #5221 to the **formal-conjectures** repository [3] (opened 31 August 2026; not merged as of 9 October 2026).

At the commit of `formal-conjectures` we worked against, the statement was still tagged `research open`, and that is why we took it up. We found the argument, completed the Lean formalization and checked it before we searched for prior work. Only then did we learn of the two works above. Our argument was therefore found independently of both.

**What this note contributes.** Given the above, our contribution is modest:

1. A second, independent proof. It uses only pointwise lower bounds on complement degrees, the inequalities  $(x + y)^2 \geq x^2 + 2xy$  and  $c^2 \geq tc$  (for  $c \geq t \geq 0$ ), and one double count. No Cauchy–Schwarz or AM–GM step is needed. All of the arithmetic is collected into a single polynomial inequality  $P(k, t) > 0$  (Lemma 5).
2. A compact formalization in Lean 4 with Mathlib (about 220 non-blank lines including the three arithmetic lemmas), which fills the `sorry` in the existing `formal-conjectures` file without changing the statement.

**Size comparison.** Pull request #5221 describes its proof as roughly 330 lines, including nine private helper lemmas [3]. Our formalization is about 220 non-blank lines with three helper lemmas, so it is roughly a third shorter. Line counts depend on style, and we have not compared the two proofs line by line. We have not seen the length of Henry’s informal proof [2], so we make no size comparison with it.

## 2 Statement

All graphs are finite and simple. For a graph  $G$  with vertex set  $V$  and  $n = |V|$  we write  $N(v)$  for the (open) neighbourhood of  $v$ ,  $\deg v = |N(v)|$ ,  $\alpha(G)$  for the independence number, and  $\overline{G}$  for the complement, so that  $\deg_{\overline{G}}(v) = n - 1 - \deg v$ .

**Definition 1.** For  $v \in V$ , let  $l(v) = \alpha(G[N(v)])$  be the independence number of the subgraph induced by the neighbourhood of  $v$  (with  $l(v) = 0$  if  $N(v) = \emptyset$ ). The *length* of a graph  $H$  is the Euclidean norm of its degree sequence,  $\text{length}(H) = (\sum_v \deg_H(v)^2)^{1/2}$ .

**Theorem 2** (WOWII Conjecture 100). *Let  $G$  be a connected graph with at least two vertices. Then*

$$\alpha(G) \leq \left\lceil \frac{\max_v l(v) + \frac{1}{2} \text{length}(\overline{G})}{2} \right\rceil.$$

*Remark 3.* The hypothesis  $n \geq 2$  is needed. For the one-vertex graph  $K_1$  we have  $\alpha = 1$ ,  $l(v) = 0$  and  $\text{length}(\overline{G}) = 0$ , so the right-hand side is 0. The WOWII text says only “simple connected graph”; the `formal-conjectures` statement assumes the vertex type is nontrivial, and we do the same.

## 3 The proof

Fix  $G$  as in Theorem 2. Throughout we use the following notation.

- $I$  is a maximum independent set,  $a = |I| = \alpha(G)$ .
- $O = V \setminus I$  and  $m = |O|$ .

- $K = \max_v l(v)$ .
- $y(w) = |N(w) \cap I|$  for  $w \in O$ .
- $E$  = the number of edges between  $I$  and  $O$ .
- $S = \sum_{v \in V} \deg_{\overline{G}}(v)^2$ , so  $\text{length}(\overline{G}) = \sqrt{S}$ .
- $D = 2a - 2 - K$ .

### 3.1 Reduction

Since  $a$  is an integer,  $\lceil x \rceil \geq a$  holds if and only if  $x > a - 1$ . With  $x = (K + \frac{1}{2}\sqrt{S})/2$  this is equivalent to

$$\sqrt{S} > 2D = 2(2a - 2 - K). \quad (1)$$

If  $D < 0$  then (1) is immediate because  $\sqrt{S} \geq 0$ . So from now on we assume  $D \geq 0$ , and it suffices to prove

$$S > 4D^2. \quad (2)$$

### 3.2 Elementary facts

**Lemma 4.** *With the notation above:*

(1)  $y(w) \leq l(w) \leq K$  for every  $w \in O$ .

(2)  $a \leq E \leq Km$ .

(3) For  $u \in I$ ,  $\deg_{\overline{G}}(u) \geq (a-1) + (m - \deg u)$ , and  $m - \deg u \geq 0$ . For  $w \in O$ ,  $\deg_{\overline{G}}(w) \geq a - y(w)$ .

*Proof.* (1) The set  $N(w) \cap I$  is independent in  $G$  (it is a subset of  $I$ ) and is contained in  $N(w)$ , so it is an independent set of  $G[N(w)]$ . Hence  $y(w) \leq \alpha(G[N(w)]) = l(w) \leq K$ .

(2) Because  $I$  is independent, every neighbour of a vertex  $u \in I$  lies in  $O$ ; hence  $\deg u = |N(u) \cap O|$  and  $E = \sum_{u \in I} \deg u$ . Since  $G$  is connected and has at least two vertices, every vertex has degree at least 1, so  $E \geq |I| = a$ . Counting the same edges from the  $O$  side gives  $E = \sum_{w \in O} y(w) \leq Km$  by (1).

(3) For  $u \in I$ , the vertices of  $I \setminus \{u\}$  are not adjacent to  $u$  in  $G$ , and neither are the vertices of  $O \setminus N(u)$ . Since  $N(u) \subseteq O$ , the latter set has  $m - \deg u \geq 0$  elements. (In fact equality holds, but we only need the inequality.) For  $w \in O$ , the  $a - y(w)$  vertices of  $I \setminus N(w)$  are non-neighbours of  $w$  distinct from  $w$ .  $\square$

Summing  $m - \deg u$  over  $u \in I$  and using (2) gives

$$\sum_{u \in I} (m - \deg u) = am - E \geq 0. \quad (3)$$

### 3.3 Case $K \geq a$

Here  $0 \leq D = 2a - 2 - K \leq a - 2$ , so  $a \geq 2$ . Keeping only the vertices of  $I$  and using Lemma 4(3),

$$S \geq \sum_{u \in I} \deg_{\overline{G}}(u)^2 \geq a(a-1)^2.$$

It remains to check  $a(a-1)^2 > 4(a-2)^2 \geq 4D^2$ . For  $a = 2$  this reads  $2 > 0$ ; for  $a = 3$  it reads  $12 > 4$ ; and for  $a \geq 4$  we have  $a(a-1)^2 \geq 4(a-1)^2 > 4(a-2)^2$ . This proves (2).

### 3.4 Case $K < a$

Put  $t = a - K \geq 1$ . Note  $K \geq 1$ : otherwise  $E \leq Km = 0 < a$ , contradicting Lemma 4(2).

*Contribution of  $I$ .* For  $x, y \geq 0$  we have  $(x + y)^2 \geq x^2 + 2xy$ . Applying this with  $x = a - 1$  and  $y = m - \deg u$  (Lemma 4(3)) and summing with (3),

$$\sum_{u \in I} \deg_{\overline{G}}(u)^2 \geq a(a - 1)^2 + 2(a - 1)(am - E).$$

*Contribution of  $O$ .* For  $w \in O$  write  $c_w = \deg_{\overline{G}}(w)$ . By Lemma 4(1),(3),  $c_w \geq a - y(w) \geq a - K = t > 0$ , hence  $c_w^2 \geq t c_w \geq t(a - y(w))$ . Summing over  $O$  and using  $\sum_{w \in O} y(w) = E$ ,

$$\sum_{w \in O} \deg_{\overline{G}}(w)^2 \geq t(am - E).$$

*Combining.* Adding the two bounds,

$$S \geq a(a - 1)^2 + (2a - 2 + t)(am - E).$$

By Lemma 4(2),  $am - E \geq am - Km = tm$ , and  $2a - 2 + t \geq 0$ , so

$$S \geq a(a - 1)^2 + (2a - 2 + t)tm. \quad (4)$$

Multiply (4) by  $K > 0$  and use  $Km \geq a$  (again Lemma 4(2)):

$$KS \geq Ka(a - 1)^2 + (2a - 2 + t)ta.$$

Since  $D = 2a - 2 - K = a - 2 + t$ , we get

$$KS - 4KD^2 \geq Ka(a - 1)^2 + (2a - 2 + t)at - 4K(a - 2 + t)^2. \quad (5)$$

Substituting  $a = K + t$  and expanding, the right-hand side of (5) equals  $P(K, t)$ , where

$$\begin{aligned} P(k, t) = & k(k^3 - 6k^2 + 17k - 16) + tk(3k^2 - 18k + 31) \\ & + t^2(3k^2 - 13k - 2) + t^3(k + 3). \end{aligned} \quad (6)$$

(We checked this identity by symbolic expansion in SymPy. In the Lean formalization the lemma `key_poly` is stated in the unexpanded form of the right-hand side of (5), so the expansion is not a separate trusted step there.) By Lemma 5 below,  $P(K, t) > 0$ , so  $KS > 4KD^2$ , and dividing by  $K > 0$  gives (2).

### 3.5 The polynomial inequality

**Lemma 5.**  $P(k, t) > 0$  for all integers  $k \geq 1$  and  $t \geq 1$ .

*Proof.* The case  $k \geq 5$ . Every term of (6) is then positive:

- $k^3 - 6k^2 + 17k - 16$  has a single real root, near 1.61, and is positive for  $k \geq 2$  (its values at  $k = 2, 3, 4, 5$  are 2, 8, 20, 44);
- $3k^2 - 18k + 31$  has discriminant  $324 - 372 = -48 < 0$ , so it is positive for all real  $k$ ;
- $3k^2 - 13k - 2$  has its larger root near 4.48, so it is positive for  $k \geq 5$  (its value at  $k = 5$  is 8);

- $k + 3 > 0$ .

The cases  $k \leq 4$ . Here  $P(k, \cdot)$  is an explicit cubic. Writing  $t = s + 1$  with  $s \geq 0$ :

| $k$ | $P(k, t)$                 | in terms of $s = t - 1 \geq 0$ |
|-----|---------------------------|--------------------------------|
| 1   | $4t^3 - 12t^2 + 16t - 4$  | $4s^3 + 4s + 4$                |
| 2   | $5t^3 - 16t^2 + 14t + 4$  | $5s^3 - s^2 - 3s + 7$          |
| 3   | $6t^3 - 14t^2 + 12t + 24$ | $6s^3 + 4s^2 + 2s + 28$        |
| 4   | $7t^3 - 6t^2 + 28t + 80$  | $7s^3 + 15s^2 + 37s + 109$     |

For  $k = 1, 3, 4$  all coefficients in  $s$  are positive, so  $P > 0$ . For  $k = 2$ : at  $s = 0$  the value is 7, and for integers  $s \geq 1$  we have  $5s^3 - s^2 - 3s = s(5s^2 - s - 3) \geq 1$ , so the value is at least 8.  $\square$

*Remark 6.* The arithmetic margin is small. By the case analysis above, the minimum of  $P$  over positive integers is  $P(1, 1) = 4$  (the next smallest values with  $k \leq 4$  are  $P(2, 1) = 7$  and  $P(2, 2) = 8$ , and  $P(k, t) \geq 5 \cdot 44$  for  $k \geq 5$ ). This says nothing about the tightness of Theorem 2 itself, since the estimates leading to (4) and (5) lose a lot for most graphs.

### 3.6 Conclusion

In both cases (2) holds whenever  $D \geq 0$ , hence  $\sqrt{S} > 2(2a - 2 - K)$ , i.e.  $(K + \frac{1}{2}\sqrt{S})/2 > a - 1$ , and therefore  $\lceil (K + \frac{1}{2} \text{length}(\overline{G}))/2 \rceil \geq a = \alpha(G)$ . This proves Theorem 2.  $\square$

## 4 Lean formalization

### 4.1 Setting

The formalization lives in the file

FormalConjectures/WrittenOnTheWallIII/GraphConjecture100.lean

of google-deepmind/formal-conjectures [4], at commit

b3f26411f06b0b0ec1923da80b62585f23288d39,

built with Lean 4 v4.33.1 [5] and the Mathlib [6] version pinned by that commit. The theorem statement is the repository's own, unchanged; we only replaced its `sorry` by a proof and changed the category attribute from `research open` to `research solved`. The statement is:

```
theorem conjecture100 (G : SimpleGraph α) [DecidableRel G.Adj]
  (h : G.Connected) :
  let maxL := (Finset.univ.image (indepNeighborsCard G)).max' (by simp)
  (G.indepNum : ℝ) ≤
    ⌈((maxL : ℝ) + (1 / 2) * (degreeL2Norm Gc : ℝ)) / 2⌉
```

Here the vertex type  $\alpha$  carries `[Fintype α] [DecidableEq α] [Nontrivial α]`, so  $G$  has at least two vertices. `indepNeighborsCard G v` is the independence number of the neighbourhood of  $v$ , and `degreeL2Norm H` is  $\sqrt{\sum_v \deg_H(v)^2}$ . Both are defined in the repository's shared utilities. The command `#print axioms conjecture100` reports only `propext`, `Classical.choice` and `Quot.sound`, the standard axioms of Lean and Mathlib; in particular there is no `sorryAx`.

## 4.2 Structure

The proof follows Section 3 closely. Three private lemmas over  $\mathbb{Z}$  carry the arithmetic, and the main theorem does the combinatorics.

- **key\_poly** ( $k\ t : \mathbb{Z}$ ): for  $k, t \geq 1$ ,  $4k(k+2t-2)^2 < k(k+t)(k+t-1)^2 + (2(k+t)-2+t)(k+t)t$ , which is  $P(k, t) > 0$  in unexpanded form. The proof splits on  $k \leq 4$  versus  $k \geq 5$ . For  $k \leq 4$ , **interval\_cases**  $k$  produces the four cubics, each closed by **nlinarith** with nonnegative products of  $t - 1$  as hints. For  $k \geq 5$ , **nlinarith** is given products of  $k - 5 \geq 0$  and  $t \geq 0$  as hints. (So the Lean certificate for  $k \geq 5$  is a Positivstellensatz-style combination found by **nlinarith**, not literally the term-by-term argument of Lemma 5.)
- **arith\_lt**: the case  $K < a$ . From the integer hypotheses  $a \leq E \leq Km$ , the  $I$ -bound  $a(a-1)^2 + 2(a-1)(am-E) \leq X$  and the  $O$ -bound  $(a-K)(am-E) \leq Y$ , it derives  $4(2a-2-K)^2 < X+Y$ , following (4)–(5) and calling **key\_poly**.
- **arith\_ge**: the case  $a \leq K$ . From  $a(a-1)^2 \leq X$  and  $0 \leq 2a-2-K$  it derives  $4(2a-2-K)^2 < X$ .
- **conjecture100**: obtains  $I$ , defines  $y$ , proves Lemma 4 and the two sums, splits into the three cases ( $D < 0$ ,  $K < a$ ,  $K \geq a$ ), and finally passes from the integer inequality  $4D^2 < S$  to the real inequality with the ceiling.

## 4.3 Mathlib API used

The combinatorial steps reduce to a handful of Mathlib lemmas.

| Mathlib declaration                                     | Used for                                                 |
|---------------------------------------------------------|----------------------------------------------------------|
| <b>SimpleGraph.exists_isNIndepSet_indepNum</b>          | a maximum independent set $I$ , $ I  = a$                |
| <b>IsIndepSet.card_le_indepNum</b>                      | $y(w) \leq l(w)$ and $a \geq 1$                          |
| <b>Preconnected.degree_pos_of_nontrivial</b>            | every degree is $\geq 1$                                 |
| <b>neighborFinset_compl</b>                             | neighbours in $\overline{G}$ as $N(v)^c \setminus \{v\}$ |
| <b>Finset.sum_card_bipartiteAbove_eq</b>                | double counting $E$                                      |
| <b>sum_card_bipartiteBelow</b>                          |                                                          |
| <b>Int.le_ceil_iff</b> , <b>Real.lt_sqrt_of_sq_lt</b>   | the final reduction (1)                                  |
| <b>nlinarith</b> , <b>interval_cases</b> , <b>omega</b> | arithmetic                                               |

The whole proof, including the three arithmetic lemmas, occupies about 220 non-blank lines. To check it:

```
git clone https://github.com/google-deepmind/formal-conjectures
cd formal-conjectures
git checkout b3f26411f06b0b0ec1923da80b62585f23288d39
git apply ../formal-conjectures.patch
lake exe cache get
lake build FormalConjectures.WrittenOnTheWallIII.GraphConjecture100
```

## 5 Computational check

Before the proof was found, the inequality was tested exhaustively on all connected graphs with  $2 \leq n \leq 9$  vertices, generated with **geng** from nauty [7]. There are  $1 + 2 + 6 + 21 + 112 + 853 + 11117 + 261080 = 273,192$  such graphs up to isomorphism. No counterexample was found. For

this note we repeated the check with a separate script that avoids floating point: by the reduction in §3.1, the inequality holds if and only if  $4a - 4 - 2K < 0$  or  $S > (4a - 4 - 2K)^2$ , which is an integer comparison. The repeat also found no counterexample. This is evidence, not part of the proof; Graffiti.pc conjectures are by construction consistent with the program’s own database of graphs.

## 6 Discussion

We have not studied the argument of [2] in detail, beyond noting that it uses the Cauchy–Schwarz and AM–GM inequalities, so we do not attempt a detailed comparison. Our proof uses only the termwise bounds of Lemma 4(3), the two elementary inequalities  $(x + y)^2 \geq x^2 + 2xy$  and  $c^2 \geq tc$ , and the double count  $a \leq E \leq Km$ ; the price is the case analysis in Lemma 5. One practical consequence of this shape is that the formal proof needs no real analysis until the last step: every inequality before it is between integers, which `nlinarith` and `omega` handle directly.

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## References

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- [3] vatsj, *Lean proof of WOWII Conjecture 100*, pull request #5221 to google-deepmind/formal-conjectures, opened 31 August 2026, <https://github.com/google-deepmind/formal-conjectures/pull/5221>.
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